

Forecasting Target's Sales of Ice Cream Sandwiches

Group 2
Fall 2025: SCMA 669-901



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Presented by

Connor Poole

Lydia Legan

Tae il Kim



Executive Summary

Mission: Provide Target with forecasting insights that transform uncertainty into strategic advantage

Data Preparation

Analyzed 10 days of daily sales and 5 weeks of weekly aggregated sales for Target's new ice cream sandwich launch.

Forecasting Hierarchy

Constructed both daily and weekly forecasts and linked them through a simple hierarchical framework.

Model Comparison

Compared TSLM, ETS, ARIMA, and Ensemble models using a consistent train/validation split.

Recommendations & Insight

Provided practical short-term recommendations and outlined data needs for future model improvements.

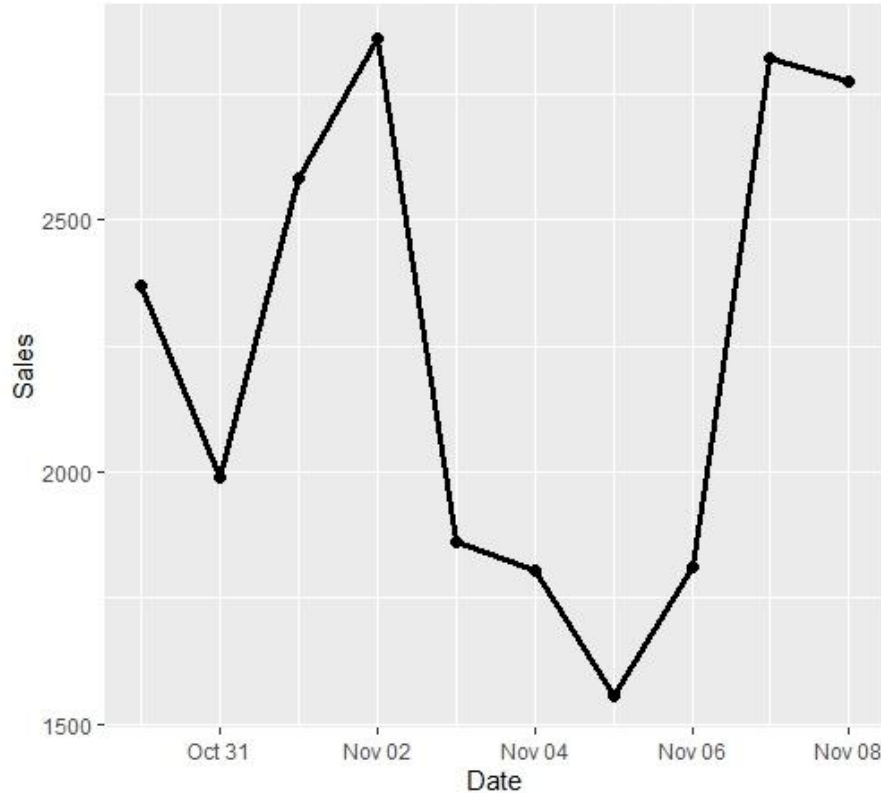
Dataset Overview & Preparation

- **Daily data:** 10 observations from Oct 30 to Nov 8
- **Weekly data:** 5 aggregated weekly totals from Oct 11 to Nov 8 representing the same product line (ice cream sandwiches)
- New product launch -> very limited history and no visible seasonality
- **Objective:** generate reliable short-term forecasts that are honest about uncertainty



Total Daily Sales for Target Product

Concatenated from 10 daily files (Oct 30 – Nov 8)

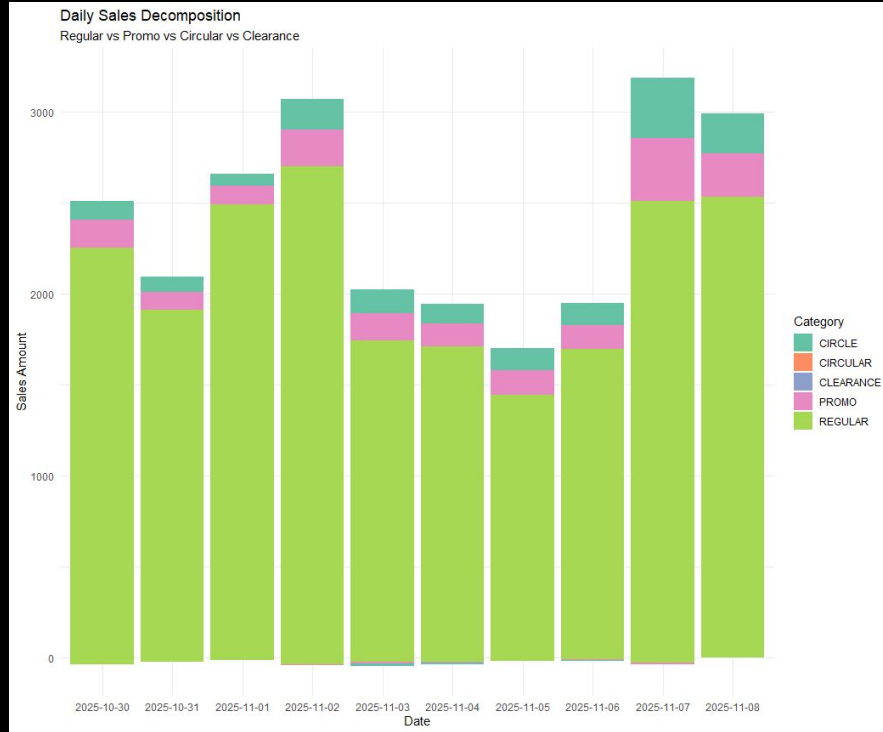


Target Products: Total Daily Sales

Daily sales fluctuate significantly during the first 10 days of the product launch, with **no clear trend or seasonal pattern**.

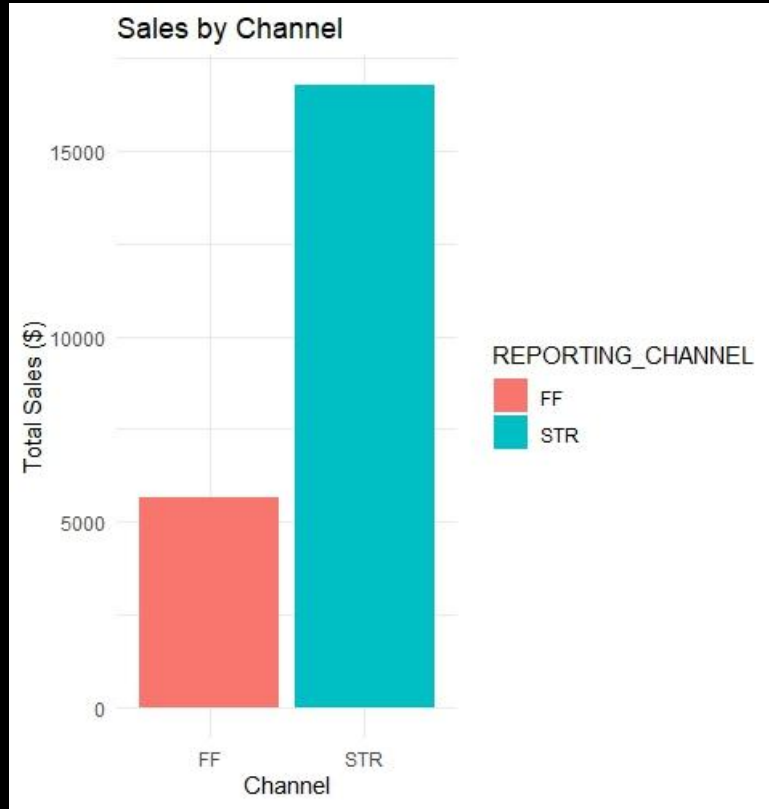
This behavior is typical for new items with limited sales history and supports the use of simple, robust forecasting models.

Early Demand Patterns



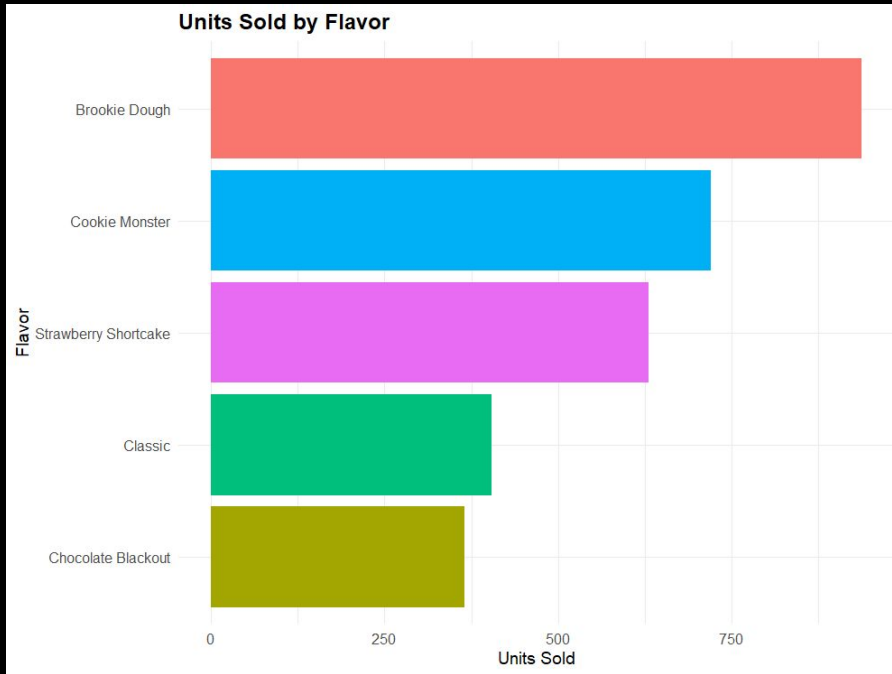
- Sales vary significantly across the first 10 days of the launch.
- Most sales come from regular-priced purchases, not promotions.
- Promo, Circle, and Clearance contributions are small but visible on certain days.
- The lack of strong promotional spikes indicates baseline demand is driving sales.
- This volatility and short time frame supports using simple, robust forecasting models.

Customer Behavior



- Most sales occurred in-store (STR) - about 75% of total dollars.
- Fulfillment channels (FF) like Drive-Up and Pickup contributed ~ 25%.
- The strong in-store share reflects how frozen novelty items are often impulse or convenience purchases.
- Fulfillment still plays a meaningful role, showing online convenience demand for frozen items.
- Understanding channel mix helps guide inventory allocation and promo strategy.

Early Demand Patterns



- Brookie Dough is the top-selling flavor, leading by a wide margin.
- Cookie Monster and Strawberry Shortcake form a strong mid-tier, showing solid performance.
- Classic and Chocolate Blackout show lower until movement compared to other flavors.
- Early flavor preferences help identify which products may need more inventory, stronger visibility, or targeted promotions.

Modeling Approach

Data Setup

- Daily data split into **training (days 1-7)** and **validation (days 8-10)**
- Weekly data aggregated from the same product line
- Tsibble structure used for both daily and weekly series
- Limited historical data -> simple models preferred

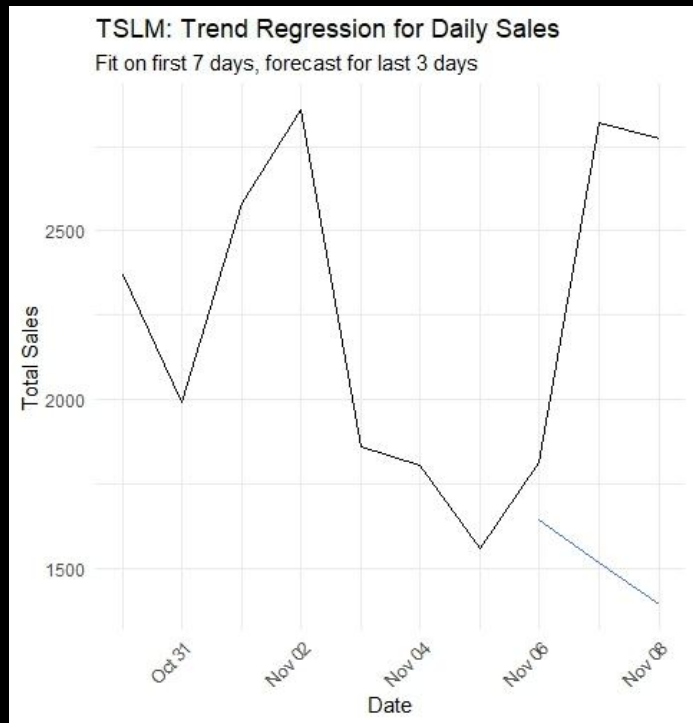
Models Tested

- **TSLM** - time series linear regression
- **ETS** - exponential smoothing
- **ARIMA(0,1,0)** - random walk model
- **Ensemble** - simple average of the above

Evaluation Procedure

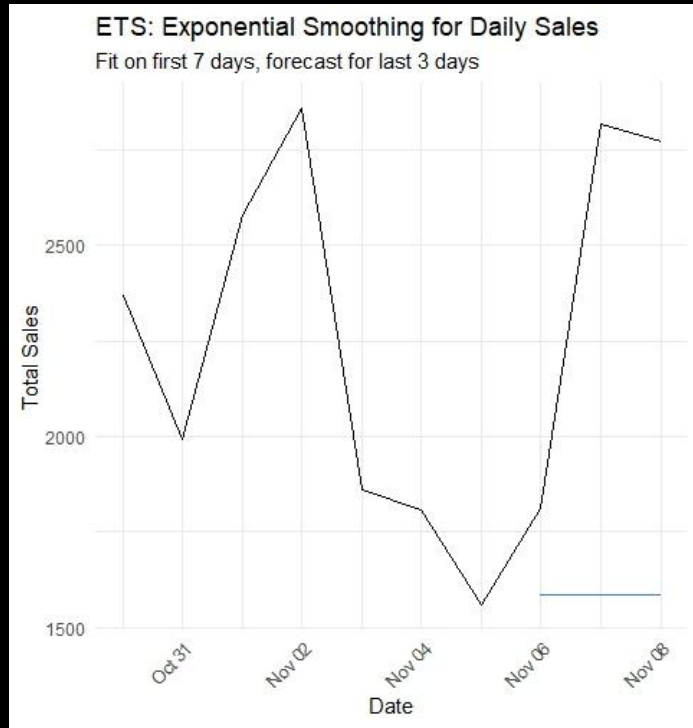
- **Stretch_tsibble()** for rolling-origin cross-validation
- Forecast horizon: **1-step ahead**
- Errors compared using **RMSE/MAE**
- Final model chosen based on **lowest validation error**

Trend Regression Model: Fit on first 7 days, forecast over the final 3 validation days



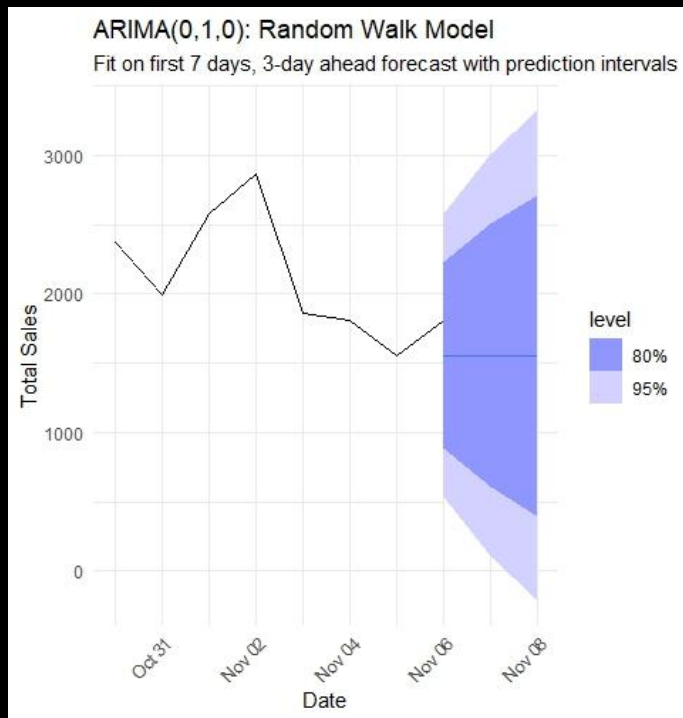
- TSLM fits sales using a simple regression: `sales ~ trend()`
- In our daily data, sales fluctuate but **do not follow** a directional trend
- Because the model assumes a steady upward or downward movement, it **forces a trend where none exists**
- This led to poor validation performance ($RMSE \sim 440$), making TSLM unsuited for this short, volatile series
- This supports the conclusion that **trend-based models are inappropriate for early-stage products with minimal history**

Exponential Smoothing Model: Exhibited overfitting on such a short series



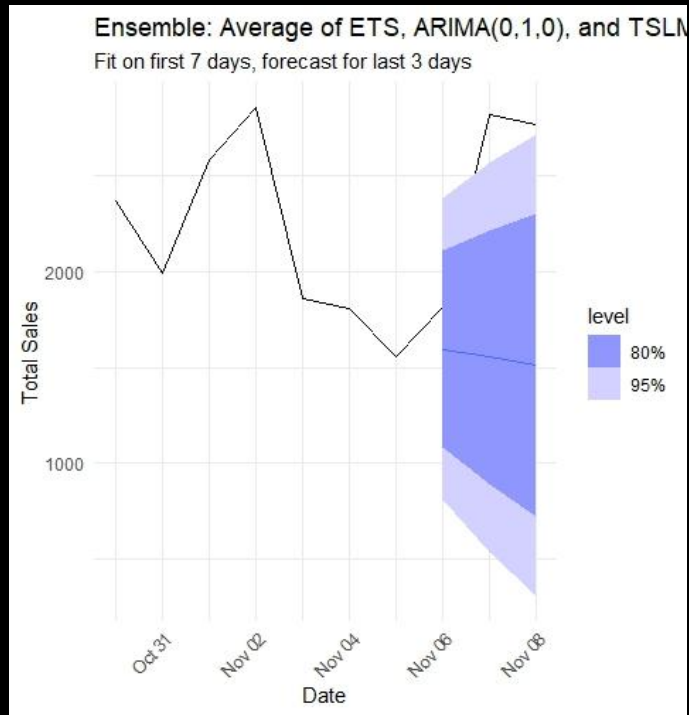
- ETS automatically estimates smoothing parameters for level, trend, and error components.
- Because daily sales are volatile and the series is extremely short, ETS **cannot stabilize** these parameters.
- Instead of capturing structure, the model **chases noise**, leading to unrealistic fitted values and poor forecasts.
- Under rolling-origin cross-validation, ETS produced the highest error values ($RMSE \sim 612$), making it the weakest model for this dataset.
- This confirms that ETS becomes **unreliable** when data history is too limited to estimate components properly.

ARIMA(0,1,0) Forecast: Best-performing model on the validation window with realistically wide prediction intervals



- The daily data showed no evidence of trend or seasonality, and the short horizon made complex models unreliable.
- ARIMA(0,1,0) also known as a **random walk**, models the next day's sales as the previous day plus random noise.
- Because this model does not estimate unnecessary parameters (unlike ETS or ARIMA with AR/MA terms), it avoids overfitting.
- Forecasts produced by ARIMA(0,1,0) are flat but accompanied by widening prediction intervals, which is appropriate for early-stage production sales.
- Under the rolling-origin cross-validation, ARIMA(0,1,0) achieved the lowest RMSE (~180) and MAE, making it the **most reliable model** for short-term forecasting on this dataset.

Ensemble Forecast: Simple average of ETS, ARIMA, and TSLM forecasts; performance was dragged down by ETS



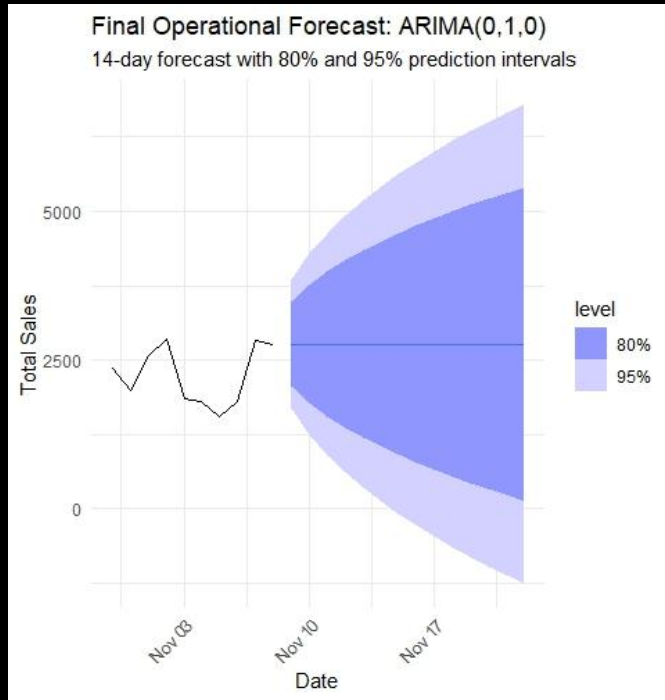
- Ensemble methods often improve accuracy by combining strengths of different models.
- In this case, the ensemble averages the output from ETS, TSLM, and ARIMA(0,1,0), each contributing equally.
- However, ETS severely overfits the short daily series, and TSLM imposes a trend that doesn't exist.
- These weaker components pull the ensemble's predictions away from the more accurate ARIMA forecasts.
- As a result, the ensemble delivered an **RMSE of ~ 401**, better than ETS or TSLM individually but significantly worse than ARIMA(0,1,0).

Comparing Models

Model	RMSE	Interpretation
ARIMA(0,1,0)	180	Best-performing model; stable and robust
Ensemble	401	Hurt by poor ETS/TSLM components
TSLM	440	Trend assumption not supported
ETS	612	Severe overfitting; unstable parameters

- **ARIMA(0,1,0)**
 - Most accurate model
 - Handles noise well
 - Forecasts remain steady and realistic
 - Ideal for early-stage product data
- **TSLM**
 - Forces a linear trend
 - No meaningful trend present
 - Overfits noise
- **ETS**
 - Requires longer histories
 - Smoothing parameters unstable
 - Worst performance
- **Ensemble**
 - Good idea in theory
 - Dragged down by poor ETS/TSLM components
 - Underperforms ARIMA

Final Operational Forecast: 14-day ARIMA(0,1,0) forecast with 80% and 95% prediction intervals for short-term planning

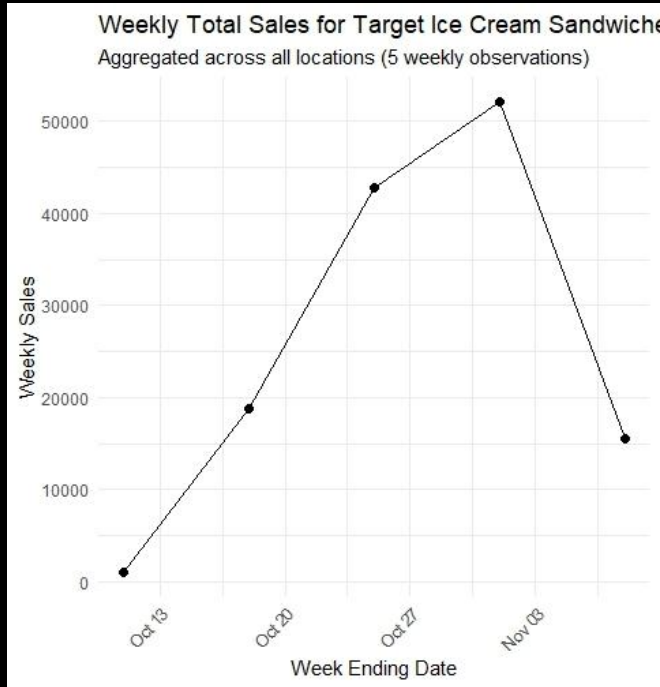


- ARIMA(0,1,0) was fit to all 10 daily observations.
- The model generates a flat mean forecast centered on the latest sales level.
- Uncertainty increases sharply across the 14-day horizon, shown by wide 80% and 95% intervals.
- This forecast reflects realistic stability for new products with volatile and limited sales history.

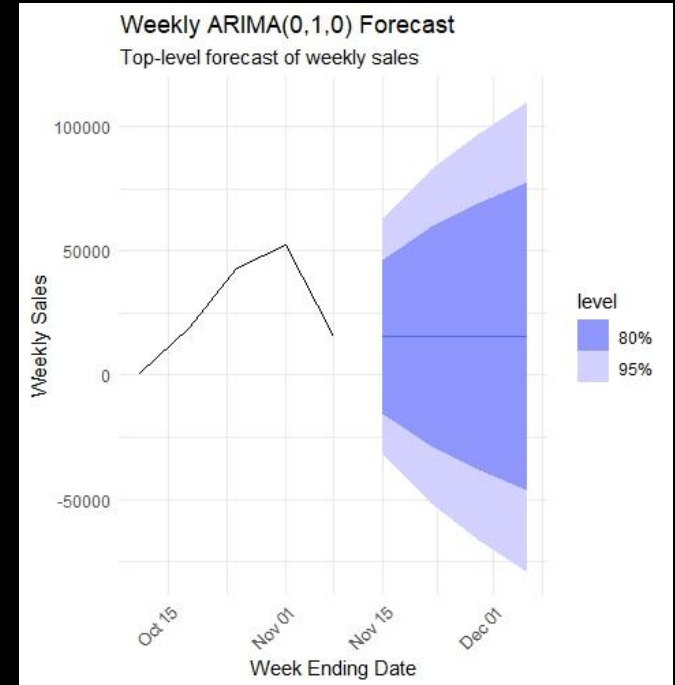
Insights:

- Forecast suggests stable demand but high risk of volatility in the short run.
- Inventory planning should assume a conservative baseline with flexibility.
- Recommend re-estimating the model after 4-8 additional weeks of data.
- Forecast intervals help Target plan buffer stock or safety inventory.

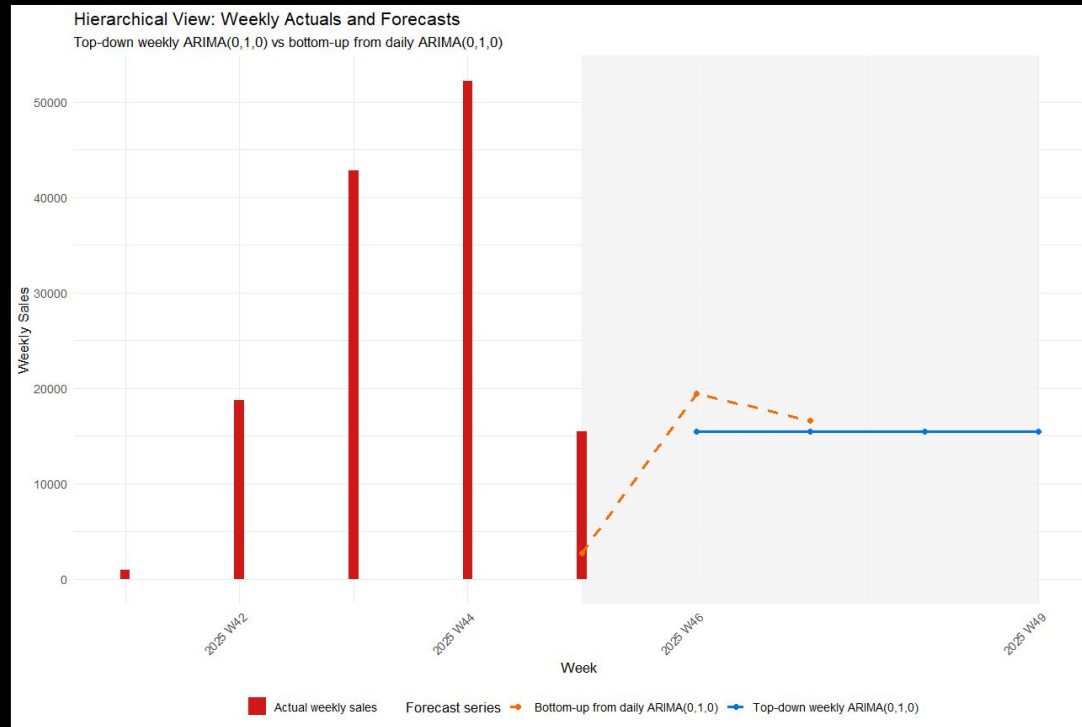
Weekly Forecast & ARIMA Forecast



- Aggregated daily sales into weekly totals to obtain a top-level view.
- Fitted a weekly ARIMA(0,1,0) model consistent with the daily random-walk assumption.
- Weekly forecasts support higher-level planning (e.g., distribution, inventory).



Hierarchical Forecast Comparison: Polished plot showing actual weekly sales (bars) plus top-down and bottom-up weekly forecasts (lines)



- Actual weekly sales highlight strong mid-launch growth followed by a drop.
- A weekly ARIMA(0,1,0) model gives a flat mean forecast with widening uncertainty.
- The bottom-up forecast, created from summed daily ARIMA predictions, captures day-level noise but converges to the same overall pattern.
- Both top-down and bottom-up approaches suggest stable demand with no detectable weekly trend, appropriate for early-stage products.

Business Recommendations for Target



- Use conservative short-term ordering given wide prediction intervals and limited history.
- Monitor sales daily and refresh forecasts as new data arrives.
- Leverage weekly forecasts for distribution and allocation decisions; rely on daily forecasts for in-store replenishment.
- Re-estimate models after 4-8 additional weeks of data, when trend/seasonality may begin to emerge.

Data at the store level:

- Add promotion flags
- Add weather data (ice cream is weather-sensitive)
- Add store-level variation (urban vs suburban)
- Track weekday effects (weekends sell more)
- Monitor out-of-stocks (common in frozen items)